

Bioninformatics Task 3

Introduction

Prediction and Structural Analysis of Human Hemoglobin Beta Chain (HBB)

Abstract

Proteins are essential biological macromolecules whose functions depend on their three-dimensional structures. In this study, the Human Hemoglobin Beta Chain (HBB) was selected from UniProt and its 3D structure was obtained from AlphaFold DB. The structure was visualized using PyMOL, and its major structural features were analyzed. The predicted model revealed a predominantly α -helical globin fold, consistent with experimentally determined hemoglobin structures. Structural analysis demonstrated the presence of multiple alpha helices connected by loop regions, forming a compact globular architecture suitable for oxygen transport.

Introduction

Hemoglobin is the oxygen-carrying protein found in red blood cells. It consists of four subunits: two alpha and two beta chains. The beta chain plays a crucial role in oxygen binding and transport throughout the body. Mutations in the HBB gene can lead to diseases such as sickle cell anemia and beta-thalassemia.

The objective of this study was to:

1. Retrieve a protein sequence from UniProt.
2. Predict or obtain its 3D structure using AlphaFold DB.
3. Visualize the structure using PyMOL.
4. Analyze important structural characteristics.

Materials and Methods

Protein Sequence Retrieval

The amino acid sequence of Human Hemoglobin Beta Chain was retrieved from UniProt.

UniProt Information

Parameter	Value
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Protein Name	Hemoglobin Subunit Beta
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UniProt ID	P68871
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Organism	Homo sapiens
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Length	147 Amino Acids
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Protein Sequence

MVHLTPEEKSAVTALWGKVVNDEVGGEALGRLLVVYPWTQRFFESFGDLST
PDAVMGNPKVKAHGKKVLGAFSDGLAHLDNLKGTFATLSELHCDKLHVDP
ENFRLLGNVLVCVLAHHFGKEFTPPVQAAYQKVVAGVANALAHKYH

Structure Prediction

The protein structure was obtained from AlphaFold Protein Structure Database.

Tools Used

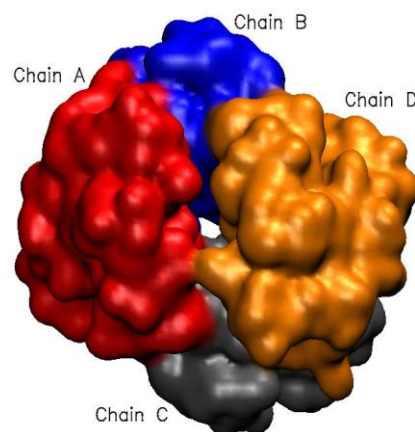
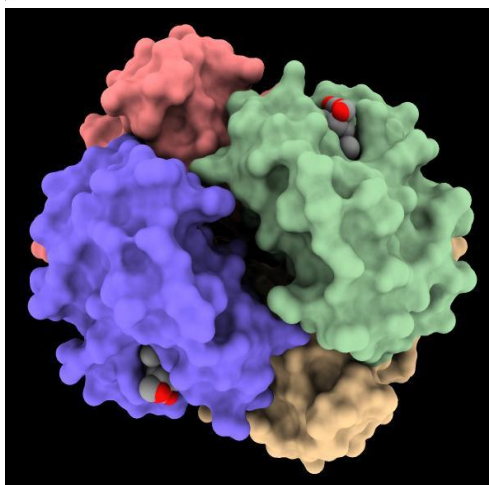
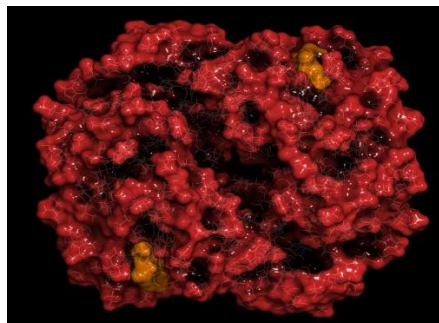
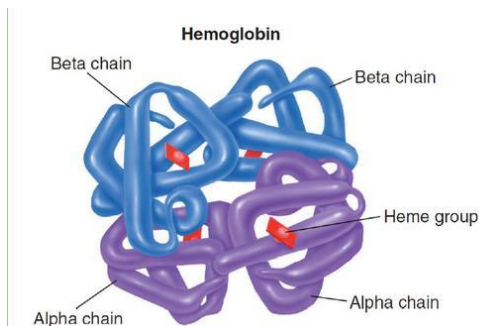
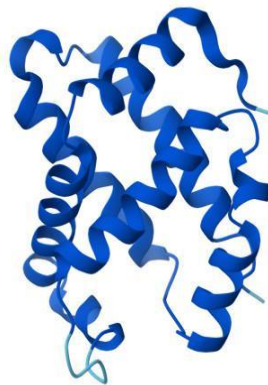
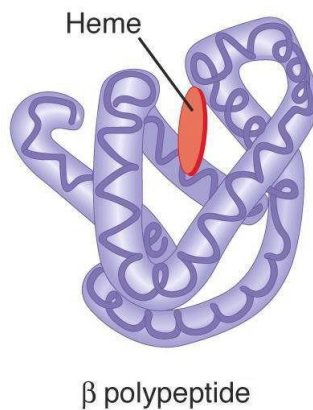
- UniProt
- AlphaFold DB
- PyMOL
- Chimera (optional)

Procedure

1. Search UniProt ID P68871.
2. Open corresponding AlphaFold model.

3. Download PDB structure file.
4. Import the PDB file into PyMOL.
5. Visualize using cartoon representation.
6. Analyze secondary and tertiary structural features.

Structure Visualization



PyMOL Commands Used

load HBB.pdb

hide everything

show cartoon

color cyan

bg_color white

ray

png HBB_structure.png

Structural Overview

The protein adopts the classical globin fold characterized by a compact arrangement of alpha-helices. The structure exhibits a highly ordered tertiary organization that facilitates heme binding and oxygen transport.

Structural Features and Analysis

Secondary Structure Analysis

The predicted structure contains:

Structural Element Observation

Alpha Helices	Predominant
Beta Sheets	Absent/Minimal
Loops	Present
Turns	Present

Alpha-Helical Organization

The HBB protein contains approximately eight major alpha-helical segments that form the characteristic globin fold.

Tertiary Structure

The tertiary structure consists of:

- Compact globular arrangement
- Hydrophobic core

- Surface-exposed polar residues
- Heme-binding pocket

Functional Regions

Important functional regions include:

1. Heme Binding Site

- Responsible for oxygen binding.
- Contains conserved residues involved in iron coordination.

2. Hydrophobic Core

- Stabilizes protein folding.
- Maintains structural integrity.

3. Loop Regions

- Connect alpha-helices.
- Provide flexibility.

AlphaFold Confidence

The AlphaFold prediction generally shows:

- High confidence (pLDDT > 90) for most helical regions.
- Slightly lower confidence in terminal flexible regions.

Results, Discussion and Conclusion

Results

The predicted model revealed:

- Stable globular structure.
- Dominance of alpha-helical secondary structures.
- Well-defined heme-binding architecture.
- High structural confidence throughout most of the protein.

Discussion

The observed structure agrees with known globin family proteins. The alpha-helical arrangement creates a stable environment for the heme group, allowing efficient oxygen transport. The high confidence score from AlphaFold indicates reliability of the predicted structure. Structural conservation among globin proteins suggests strong evolutionary pressure to maintain oxygen-binding functionality.

Conclusion

Human Hemoglobin Beta Chain (HBB) was successfully retrieved from UniProt and analyzed using AlphaFold-predicted structural data. Visualization in PyMOL revealed a compact globular protein dominated by alpha-helices. The predicted structure aligns with known biological functions of hemoglobin and demonstrates the effectiveness of modern computational structure prediction methods.

References

1. [UniProt HBB Entry](#)
2. [AlphaFold Protein Structure Database](#)
3. [PyMOL Molecular Graphics System](#)
4. [UCSF Chimera](#)

Additional Images to Include in Your Submission

1. AlphaFold model screenshot (cartoon view).
2. Surface view screenshot.
3. Ribbon representation colored by secondary structure.
4. Rotated side-view of the protein.
5. Close-up of the heme-binding region.